Advanced Mathematical Techniques Problem Set 1
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The purpose of this worksheet is to give you some more practice with convergence tests, using the integral comparison test to find estimates for the values of convergent series and particular sums of portions of divergent series, and using Taylor series to determine the value of various derivatives of functions. Please show all of your work on each problem and only use a calculator or Mathematica when specifically asked for decimal approximations.

1. We showed in class that the harmonic sequence defined by $H_n = \sum_{k=1}^{n} \frac{1}{k}$ diverges, but it doesn't diverge in a manner that might popularly be described as *quickly*. The purpose of this problem is to explore the rate at which this series diverges. We'll start with the sum of the first thousand terms, $H_{1000} \approx 7.48547$.
   (a) Use the integral comparison inequality to bound the value of $H_{10^{12}}$. How far apart are your bounds?

   (b) If one term of the harmonic series was added each second since the beginning of the universe, 13.7 billion years ago, use the integral comparison inequality to bound the resulting sum. Use $3.155 \times 10^7 \text{s}$ for one year. How far apart are your bounds?

   (c) If one *trillion* terms of the harmonic series were added each second since the beginning of the universe, 13.7 billion years ago, use the integral comparison inequality to bound the resulting sum. Use $3.155 \times 10^7 \text{s}$ for one year. How far apart are your bounds?

   Your results for the previous parts are relatively small numbers in comparison with the number of terms added. Despite that fact, we have shown that the harmonic series *diverges*. This means that the sum will eventually exceed *any* given number.
   (d) Use the integral comparison inequality to estimate the number of terms that must be added for the sum to exceed 100. Give your answer in scientific notation with mantissa rounded to the nearest thousandth.

   (e) Use the integral comparison inequality to estimate the number of terms that must be added for the sum to exceed 1000. Give your answer in scientific notation with mantissa rounded to the nearest thousandth.

   (f) Explain what is meant by the statement that the harmonic series diverges, but it does not do so *quickly*.

We established in class that there *is no* slowest diverging series because the terms in *any* divergent series can be divided by the partial sum of its terms to obtain another divergent series. Applying this process to the harmonic series leads to what we will call the *glacial* sequence, $G_n = \sum_{k=2}^{n} \frac{1}{k \ln k}$. Given $G_{1000} = 2.7274$,

(g) use the integral comparison inequality to bound the value of $G_{10^{12}}$. How far apart are your bounds?

(h) use the integral comparison inequality to bound the sum obtained by adding one term per second since the beginning of the universe. How far apart are your bounds?

(i) use the integral comparison inequality to bound the sum obtained by adding a *trillion* terms per second since the beginning of the universe. How far apart are your bounds?

(j) use the integral comparison inequality to determine *approximately* how many terms would need to be added in order for the sum to exceed 100. Give your answer in the form $10^{a \times 10^n}$, with n an integer and a rounded to the nearest thousandth. Explain why it would be very difficult to determine the answer to this question in scientific notation with mantissa rounded to the nearest thousandth.

2. The integral comparison inequality can also be used to estimate the value of *convergent* series.

(a) Given that $\sum_{k=2}^{100} \frac{\ln^3 k}{k^2} = 4.055834347...$, show that $\sum_{k=2}^{\infty} \frac{\ln^3 k}{k^2}$ converges and bound its value. How far apart are your bounds?

(b) Given that $\sum_{k=2}^{100} \frac{1}{k \ln^2 k} = 1.89283...$, show that $\sum_{k=2}^{\infty} \frac{1}{k \ln^2 k}$ converges and bound its value. How far apart are your bounds?

(c) How significant are the contributions from $k > 100$ to the complete infinite sums in parts (a) and (b)? Explain whether or not you think that the contributions coming from the integral comparison inequality are important and your reasons for your response.

3. Given the function $f(x) = \int_0^x \frac{1-\cos\left(2t^2\right)}{t^3}\,dt$,

   (a) Write the Taylor expansion of the integrand of this integral centered at $t = 0$.

   (b) Integrate your expression from part (a) term-by-term to determine the Taylor series expansion for $f(x)$ centered at 0.

   (c) Use your result from part (b) to determine $f^{(10)}(0)$. You may leave factorials in your answer if you like.

   (d) Use your result from part (b) to determine $f^{(20)}(0)$. You may leave factorials in your answer if you like.

4. Given the function $g(x) = x^2 \int_0^x \frac{\ln\left(5+2t^3\right)-\ln 5}{t^3}\,dt$,

   (a) Write the Taylor expansion of the integrand of this integral centered at $t = 0$.

   (b) Integrate your expression from part (a) term-by-term, then multiply your result by x^2 to determine the Taylor series expansion for $g(x)$ centered at 0.

   (c) Use your result from part (b) to determine $g^{(11)}(0)$. You may leave factorials in your answer if you like.

   (d) Use your result from part (b) to determine $g^{(18)}(0)$. You may leave factorials in your answer if you like.

5. Determine the first four nonzero terms in the Maclaurin expansion of the function $h(x) = \frac{x}{1-x+x^2}$ and give the radius of convergence of this expansion.

6. For this problem, consider the sequence $\{a_k\}_{k=1}^{\infty}$ of positive numbers and the sequence $\{S_n\}_{n=1}^{\infty}$ defined by $S_n = \sum_{k=1}^{n} a_k$. We will assume $\{S_n\}_{n=1}^{\infty}$ diverges.

(a) Explain why the sequence $\{S_n\}_{n=1}^{\infty}$ is monotonic increasing.

(b) Explain why the series $\sum_{n=N+1}^{N+P} \frac{a_n}{S_n}$ is strictly greater than $\frac{1}{S_{N+P}} \sum_{n=N+1}^{N+P} a_n$ for any natural number *N* and *all* natural numbers $P > 1$. Then, re-express the remaining sum in terms of elements of the sequence *S* of partial sums and simplify your result as much as possible. This is exactly as we did in class.

(c) Explain why the result of part (b) implies the series $\sum_{n=1}^{\infty} \frac{a_n}{S_n}$ *also* diverges.

(d) The Harmonic sequence has $a_k = 1/k$. Find an approximation for S_n in this situation that is valid for large *n* (note that this result follows immediately from your analysis of problem 1) and use it to write the divergent series resulting from part (c). What will happen if you do this *again* with the resulting series?

(e) Explain how your analysis in part (d) implies that there *is no* 'slowest diverging series'.

We have seen that dividing the terms of the sequence $\{a_k\}_{k=1}^{\infty}$ by the terms in the sequence $\{S_n\}_{n=1}^{\infty}$ of partial sums is *not* enough to cause the resulting series to converge, i.e. $\sum_{n=1}^{\infty} \frac{a_n}{S_n}$ *also* diverges. What happens if we instead divide by S_n^2?

(f) Explain why $\frac{a_n}{S_n^2} = \frac{S_n - S_{n-1}}{S_n^2} < \frac{1}{S_{n-1}} - \frac{1}{S_n}$.

(g) Use your result to bound the series tail $\sum_{n=N+1}^{N+P} \frac{a_n}{S_n^2}$ by a simple expression containing two terms. What happens to this expression as *P* grows without bound? Can you use this to establish anything about the behavior of the series $\sum_{k=1}^{\infty} \frac{a_k}{S_k^2}$?

(h) What series do you get if you treat the Harmonic sequence in this way?